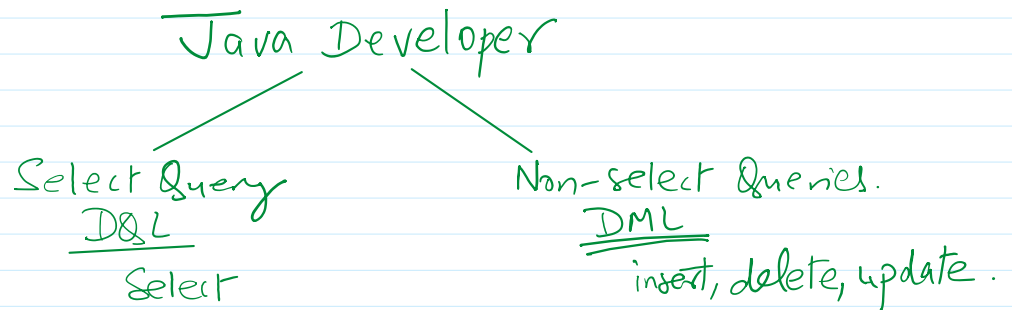


Types of SQL commands

17 May 2026 11:27

1. DDL(Data Definition Language) Commands: Create table, alter table, drop table etc.
2. DML(Data Manipulation Language) Commands: insert, delete, update
3. DQL(Data Query Language) Commands: Select
4. DCL(Data Control Language) Commands: alter password, grant access etc.
5. DAC(Data Administrative Commands: start audit, stop audit
6. TCC(Transaction Control Commands): commit, rollback, savepoint etc.



① To create a table.

create table movies (mno Number, mname Varchar2(100)....);

② To Delete / Drop table.

DROP table movies;

③ To insert rows into table

insert into movies value (1, 'Bahubali', 'Prabhas', 'Anushka');
insert into movies value (2, 'Spiderman', 'Tobbie', 'Amy');

④ Update data:

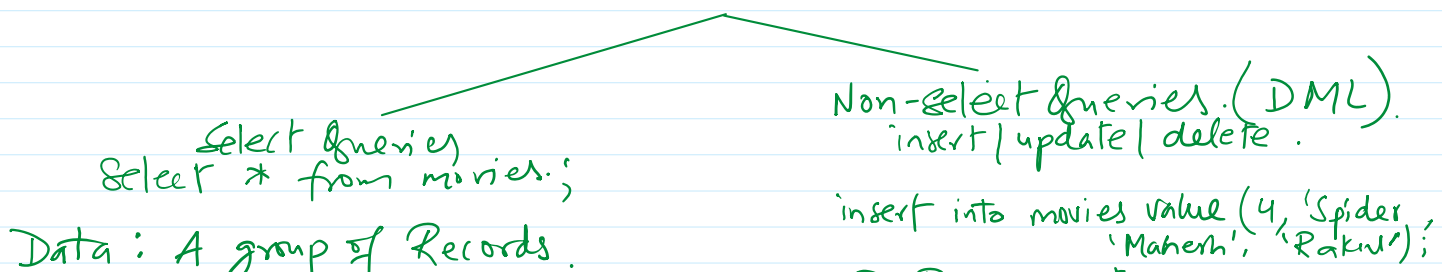
update movies set heroine = 'Tamannaah'
where mname = 'Bahubali';

⑤ To delete a row.

delete from movies where mno=2;

⑥ To select data;

select * from movies;



Data: A group of Records.

In JDBC it is represented as ResultSet.

insert into movies value (4, 'Spider', 'Mahesh', 'Rakul');

① Row create

delete from movies where mno=4;

① Row deleted

Result is number of error value.

java.sql.

Connection con =

Statement st = con.createStatement();
↑
Connection object.

SQL Query

3 - methods -

- ① executeQuery()
- ② executeUpdate()
- ③ execute()

① executeQuery()? To execute select queries.

String query = "select * from movies";

prototype: group of records $\Rightarrow$ ResultSet object.

public ResultSet executeQuery (String sql) throws SQLException.

e.g. ResultSet rs = st.executeQuery(query);

② executeUpdate(): It is applicable for non-select queries.
DML (insert / delete / update).

String sql = "delete from movies where mno > 10"

③ Row deleted.

prototype:

public int executeUpdate (String sqlQuery) throws SQLException.

e.g.:- int x = st.executeUpdate(sql);
System.out.println("The number of movies deleted: " + x);

② execute() $\Rightarrow$ both select & non-select operations.
* If we don't know the ^{Type of} query at the beginning.

prototype:-
public boolean execute(String sql) throws SQLException.

↳ true $\Rightarrow$ select query is used.

false $\Rightarrow$ non-select query

boolean b = st.execute(dynamically provided query).

```
if (b) {  
    System.out.println("It is a select query");  
    ResultSet rs = st.getResultSet();  
    ...  
}
```

```
else {
```

```
    int rowCount = st.getUpdateCount();  
    ...  
}
```

```
}
```

What these functions returns:

executeQuery() $\Rightarrow$ ResultSet

executeUpdate() $\Rightarrow$ int

execute() $\Rightarrow$ boolean

true $\swarrow$ false
getResultSet() getUpdateCount()

```
package MyJDBCOne;  
import java.sql.Connection;  
import java.sql.DriverManager;  
import java.sql.ResultSet;  
import java.sql.SQLException;  
import java.sql.Statement;  
public class OracleJDBCOne {  
    public static void main(String[] args) {  
        String dbUrl = "jdbc:oracle:thin:@//localhost:1521/FREE";  
        String username = "sys as sysdba";  
        String password = "Calcutta1983";
```

```

try {
    Connection con = DriverManager.getConnection(dbUrl, username, password);
    Statement stmt = con.createStatement();
    String query = "SELECT * FROM EMPLOYEE";
    ResultSet rs = stmt.executeQuery(query);
    while (rs.next()) {
        System.out.println("-----Employee details-----");
        System.out.println(rs.getString("EMP_ID"));
        System.out.println("Name: " + rs.getString("NAME"));
        System.out.println("Department: " + rs.getString("DEPT_ID"));
        System.out.println("Basic salary: " + rs.getFloat("BASIC_SALARY"));
        System.out.println("-----");
    }
} catch (SQLException e) {
    e.printStackTrace();
    System.err.println("Connection not established");
}
}
}

```

-----Employee details-----

E001

Name: Pratik Jain

Department: D001

Basic salary: 56000.0

-----Employee details-----

E002

Name: Naveen Singh

Department: D001

Basic salary: 67800.0

-----Employee details-----

E003

Name: Sangita Mishra

Department: D004

Basic salary: 45700.0

-----Employee details-----

E004

Name: Ajay Dewan

Department: D002

Basic salary: 34000.0

-----Employee details-----

E005

Name: Bali Agarwal

Department: D002

Basic salary: 35600.0
